

Encapsulation & Decapsulation — Guided Notes

Unit 1.1.2 | CompTIA Network+ N10-009 Obj. 1.1 | N10-008 1.1

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

When data travels down the OSI model on its way out of a device, each layer adds its own _____. This process is called _____. At the destination, each layer _____ its own header in reverse — a process called _____.

Question 2

Match each header type to the OSI layer that adds it:

Port numbers and TCP/UDP info → Layer _____

Source and destination IP addresses → Layer _____

MAC addresses and CRC trailer → Layer _____

Question 3

TCP establishes a connection using a _____ handshake. The three steps are: _____ (sender initiates), _____ (receiver acknowledges), and _____ (sender confirms).

Question 4

Explain the key difference between TCP and UDP. Give one real-world application that uses each protocol and explain why.

Question 5

List the purpose of three TCP flags:

SYN: _____

FIN: _____

RST: _____

Question 6

At Layer 2, the Data Link layer wraps a packet into a _____. It adds a _____ with MAC addresses and a _____ containing a CRC value called the _____. If the CRC check fails at the receiving end, the frame is _____.

Question 7

The MTU (Maximum Transmission Unit) on standard Ethernet is _____ bytes. If data exceeds this limit, it gets _____ into smaller pieces. Why can VPN tunnels cause MTU-related connectivity problems?

Question 8 — Real World Application

REAL WORLD: You open Wireshark on a machine and capture traffic while loading a webpage. Describe what a normal TCP connection setup looks like in the capture. What three-packet sequence would you see at the start, and what flags would each packet have? What would it mean if you saw hundreds of SYN packets but no SYN-ACK responses?

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/1.1.2-encapsulation/encapsulation-answer-key.pdf
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